FYS5429/9429 April 16

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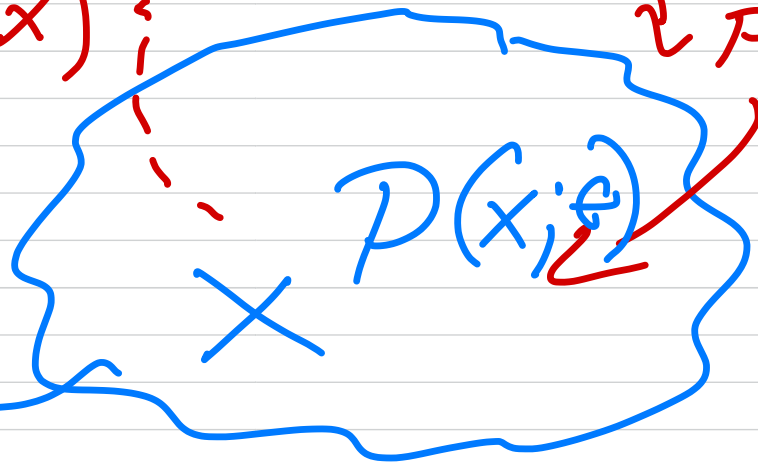
$$\mathcal{L}(X;\theta) = \pm \log P(X;\theta)$$

$$P(X;\theta) = \prod_{x_i \in X} P(x_i;\theta)$$

$h_j \in H$ — Latent/hidden space

$p(h|x)$

$p(x|h)$



$\tilde{p}(x)$ — Data set

$$\hat{\theta} = \underset{\theta \in \mathbb{R}^p}{\arg\max} \log P(x; \theta)$$

$$P(x_i; \theta) = \sum_{h_j \in H} P(x_i, h_j; \theta)$$

marginal probability

$$P(x_i, h_j; \theta) = \frac{f(x_i, h_j; \theta)}{Z(\theta)}$$

$$Z(\theta) = \sum_{x_i, h_j} f(x_i, h_j; \theta)$$

conditional probability

$$p(x|h) = p(h|x)\,p(x)$$

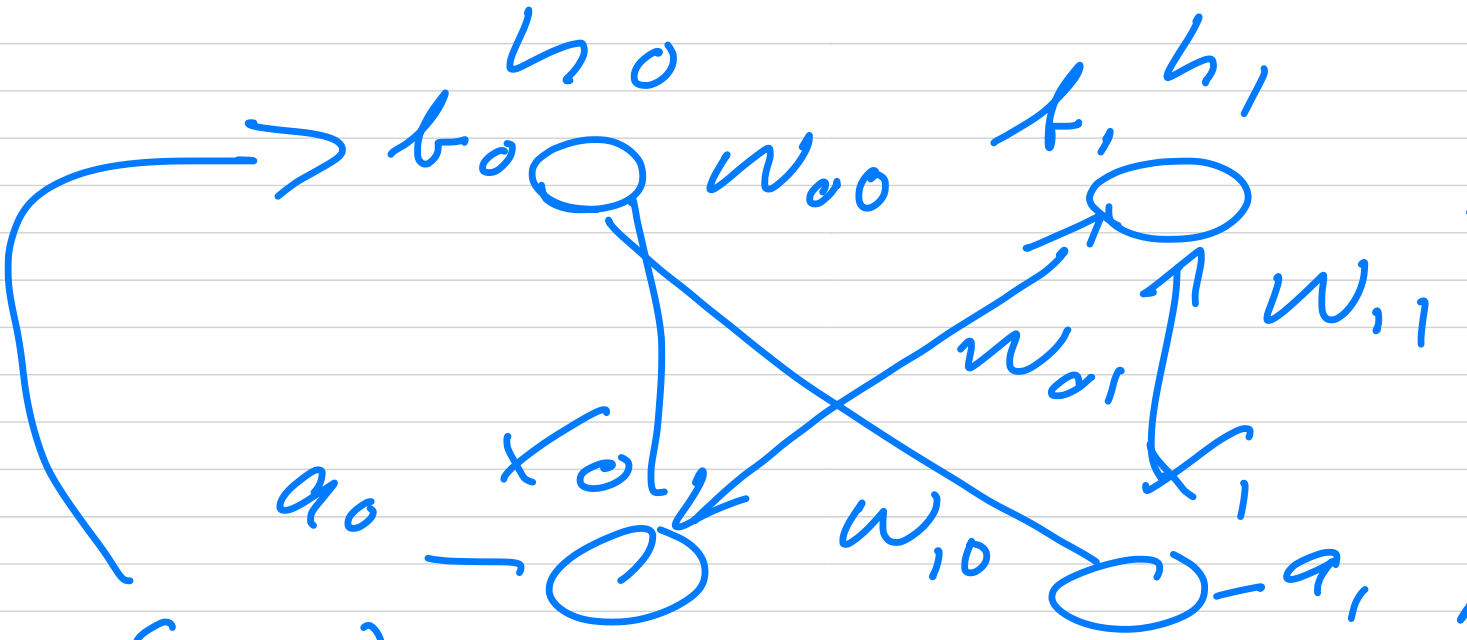
$$p(x_i, h_j; \theta) \Rightarrow p(x, h; \theta)$$

$$p(x, h; \theta) = \frac{f(x, h; \theta)}{Z(\theta)}$$

Model in $f(x, h; \theta)$

simplest case : Binary - Binary Restricted Boltzmann Machine (RBM)

$x_i = \{0,1\} \wedge h_j = \{0,1\}$



2^M

$M = 2$

2^N

$N = 2$

$\{0,0\}, \{0,1\}, \{1,0\} \quad \{1,1\}$

$$Z(\theta) = \sum_{xh} f(x, h; \theta)$$

$$= \sum_{x_0 = \{0,1\}} \sum_{x_1 = \{0,1\}} \sum_{h_0 = \{0,1\}} \sum_{h_1 = \{0,1\}} f(x_i, h_j)$$

Binary - Binary Model

$$f(x, h; \theta) = e^{-(a^T x + b^T h + x^T W h)} \quad \Big| \; \theta = \{a, b, W\}$$

$$a^T x + b^T h + x^T W h =$$

$$\sum_{i=0}^{N-1} a_i x_i + \sum_{j=0}^{M-1} b_i h_j + \sum_{ij} x_i w_{ij} h_j$$

$$\nabla_\theta \log P(x;\theta)$$

$$= \sum_{x \in X} \nabla_\theta \left[\log f(x;\theta) \right]_{DATA}$$

$$- \mathbb{E}\left[\nabla_\theta \log f(x;\theta) \right]_{MODEL} = 0$$

$$\sum_{x \in X} p(x;\theta) \nabla_\theta \log f(x;\theta)$$

$$\simeq \frac{1}{MCS} \sum_{i=1}^{MCS} \nabla_\theta \log f(x_i;\theta)$$

The last equation is evaluated using $(MC)^2$ with Gibbs sampling.

$$P_{BB}(x, h; \theta) = \frac{1}{Z_{BB}(\theta)} e^{-\left(\sum_i a_i x_i + \sum_j b_j h_j + \sum_{ij} x_i w_{ij} h_j\right)}$$

$$= \frac{1}{Z_{BB}(\theta)} e^{-\left(a^T x + b^T h + x^T W h\right)}$$

$$P_{BB}(x;\theta) = \sum_h P_{BB}(x,h;\theta)$$

$$= \frac{1}{Z_{BB}} \sum_h e^{-(a^T x + b^T h + x^T W h)}$$

$$= \frac{1}{Z_{BB}} e^{-a^T x} \sum_h e^{-\sum_j (b_j + x^T W_{*j}) h_j}$$

$$= \frac{1}{Z_{BB}} e^{-a^T x} \sum_h \prod_{j=0}^{M-1} e^{(b_j + x^T W_{*j}) h_j}$$

$$= \frac{1}{Z_{BB}} e^{-a^T x_{M-1}} \prod_{j=0} \sum_{h_j} e^{-(b_j + x^T W_{*j}) h_j}$$

$$h_j = \{0, 1\}$$

$$= \frac{1}{Z_{BB}(\theta)} e^{-a^T x_{M-1}} \prod_{j=0} \left(1 + e^{-(b_j + x^T W_{*j})}\right)$$

$$P_{BB}(h;\theta) = \frac{1}{Z_{BB}} e^{-b^T h} \times$$

$$\prod_{i=0}^{N-1} \left(1 + e^{-(a_i + x_i w_i + h)}\right)$$

RBM with probabilities for

h_i on or off

on $h_j = 1$ V off $h_j = 0$

conditional probability

$$P_{BB}(h_j = 1 \mid x) = \frac{1}{1 + e^{-(b_j + x^T w_{*j})}}$$

$$P_{BB}(h_j = 0 \mid x) = \frac{1}{1 + e^{(b_j + x^T w_{*j})}}$$

$$P_{BB}(x_i = 1 \mid h) = \frac{1}{1 + e^{-(a_i' + w_{i*}^T h)}}$$

$$P_{BB}(x_i = 0 \mid h) = \frac{1}{1 + e^{a_i + w_{i*}^T h}}$$

$$\nabla_\theta \mathcal{L}(x;\theta) \qquad e^{-E(x,h;\theta)}$$

$$\nabla_\theta \log f(x,h;\theta)$$

$$\nabla_\theta \left(-\left(\sum_{i=0}^{N-1} a_i x_i + \sum_{j=0}^{M-1} b_j h_j \right. \right.$$

$$\left. \left. + \sum_{i,j} x_i W_{ij} h_j \right) \right)$$

$$\nabla_a \log f$$

$$\frac{\partial \log f}{\partial a_i} = -x_i \wedge \frac{\partial \log f}{\partial b_j} = -h_j$$

$$\frac{\partial \log f}{\partial w_{ij}} = -x_i' h_j'$$

$(MC)^2$: $\mathbb{E}[f(x)] =$

$$\int_{x \in D} dx \, p(x) f(x)$$

$$\simeq \frac{1}{M} \sum_{i=1}^{M} f(x_i)$$

Markov-Chain

$$P(x_i, t) = \sum_j W(j \Rightarrow i) P(x_j, t-1)$$

vector-Matrix form

$$P(t) = W P(t-1)$$

Transition probability

Equilibrium

$$p(t) = p(t-1) = P$$

$$P = WP$$

our model

stochastic matrix

$$\sum_j W_{ij} = 1$$

$$\sum_i P_i = 1$$

$$W(j \to i) = \underbrace{T(j \to i)}_{\text{probab of making a transition}} A(j \to i)$$

prob of accepting move

$$P \to P_i$$

Detailed balance

$$\boxed{\frac{P_i}{P_j}} = \frac{T(j \to i)\, A(j \to i)}{T(i \to j)\, A(i \to j)}$$

Known

Metropolis algo

$$A(j \to i) = \mathrm{Min}\left(1, \frac{P_i\, T(i \to j)}{P_j\, T(j \to i)}\right)$$